Introduction to Auction Theory

Econ 177

Introduction: A Historical Perspective

- Herodotus reports that auctions were used in Babylon as early as 500 B.C.
- 193 A.D. the Praetorian Guard sold the Roman Empire by means of an auction
- Wide array of commodities sold by means of auction
  - Tobacco, fish, fresh flowers
  - Bond issues by public utilities and long-term U.S. Treasury securities
  - Total value measured in trillions!
- Facilitate transfer of assets from public to private hands
  - Industrial enterprises in Eastern Europe and former Soviet Union
  - Rights to harvest natural resources such as oil and timber
  - Rights to use the electromagnetic spectrum for communications
  - Since 1994 the FCC has conducted 87 spectrum auctions with combined revenue of $60 billion
  - IPOs — Google IPO in 2004 raised $1.67 billion

Common Auction Formats

- **Open ascending price or English auction**
  - Auctioneer begins by calling out a low price and raises it in small increments provided there are two or more active bidders
  - Auction ends when there is only one remaining bidder
- **Open descending price or Dutch auction**
  - Descending counterpart to English auction
  - Not commonly employed, but it is of conceptual interest

Common Auction Formats

- **First-Price, Sealed Bid Auction**
  - Bidders submit bids in sealed envelopes
  - At a pre-determined time, auctioneer opens all envelopes and ranks bids
  - Highest bidder obtains object and pays his bid amount
- **Second-Price, Sealed Bid Auction**
  - Bidders submit bids in sealed envelopes
  - At a pre-determined time, auctioneer opens all envelopes and ranks bids
  - Highest bidder obtains object and pays second highest bid amount

Common Auction Formats

- **All-Pay Auction**
  - Bidders submit bids (open or closed)
  - At a pre-determined time, auctioneer opens all envelopes and ranks bids
  - Every bidder pays what they bid regardless of whether or not they have the highest bid
  - Highest bidder obtains object
  - Examples:
    - Penny Auctions (Quibids, Swoopo, Bidcactus, Beezid, Ubid etc.)
    - Elections, almost any kind of contest or sports event, research and development, wars, lobbying, queues

Valuations

- We assume bidders have a use-value or simply a "value" for the object
  - Reservation price: $U(1, w - r) = U(0, w)$
- If seller knew bidder values, she could offer object to bidder with highest value at or just below his willingness to pay
- Auctions are used because seller is unsure about values bidders attach to object being sold
  - Seller does not know the values of potential bidders
  - Bidders may know their own values, but not the values of potential opponents (or the seller)

Valuations

- **Private values paradigm**
  - Bidder knows value of object to himself but not others at time of bidding
  - Knowledge of other bidders' values would not affect how much the object is worth to a given bidder
  - Assumption most plausible when values are derived from consumption or use alone
- **Common values**
  - Value is the same for all bidders
  - Each bidder has their own imprecise estimate of the value of the object
  - Appropriate when value of object is derived from a market price unknown at time of bidding
  - Tract of land with unknown quantity of oil underground
- **Interdependent values**
  - Each bidder receives a private signal about her value
  - Each bidder's value is based on a combination of the private signals

Revenue versus Efficiency

- Two criteria for evaluating performance of an auction
  - **Revenue** or expected selling price
  - **Allocative efficiency**: Does the object get sold to the bidder who values it the most or who will make the best use of it for society?
    - Important goal in the case of privatization
    - U.S. Congress directed the FCC to choose an auction design that "puts licenses into the hands of those who value them the most" (Milgrom 2004, p.5)
- **Tradeoff between revenue and efficiency**
  - Features of auctions that increase revenue typically reduce efficiency
- Other issues in choosing a mechanism
  - Susceptibility to collusion and simplicity/transparency of auction rules

Reserve Prices

- Until now we have implicitly assumed that the seller is willing to part with the object at any positive price
- If the object has some intrinsic value to the seller, she would be foolish to sell it for less than this value
- This possibility can be avoided by specifying a **reserve price** below which the object will not be sold
- Note: A seller may want to set a reserve price even if her own value for the object is zero!

Reserve Prices

- In some real-world auctions it is announced that there is a reserve price, but the actual reserve is kept secret
  - If the reserve is not met it is simply announced that the object has not been sold
  - The seller may want to keep the reserve a secret if she plans to negotiate with the highest bidder if the reserve is not met, or to try to re-auction again later
- Role of reserves and second-chance offers

Entry Fees

- Sometimes it costs money to participate in an auction
- Entry fee is paid by participants whether they win the auction or not
- Entry fees result in negative earnings for losing bidders

Agenda

- We model auctions as a strategic game:
  - **Players:** The bidders and the seller(s)
  - **Strategies:** Bidders decide how much to bid; the seller decides the auction format and rules (the auction design), e.g. reserve price
  - **Payoffs:** Depend on the design of the auction and the bids, but typically the surplus from the trade is split in some way between the winning bidder and the seller

Agenda

- Objectives of the game theoretic study of auctions are:
  - To model and predict auction outcomes
    - How do bidders decide how much to bid?
  - To understand how changes in auction rules affect bidding behaviour and impact on revenue
  - To identify the circumstances in which auctions are an efficient mechanism for allocating scarce resources
  - To understand why theoretical predictions may fail to predict auction outcomes
  - Avoid getting ripped off!

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